ME416
Ground Vehicle Dynamics

Aerodynamics –
Drag

Fall 2015

1

Aerodynamic Drag Reduction – Why?

- Recall that we generally discuss the aerodynamic drag as:

$$F_D = C_D A_f \frac{1}{2}\rho V^2$$

where C_D is the drag coefficient, A_f is the frontal area and V is the air velocity

- The power to overcome the aerodynamic drag (or any resistive force) is:

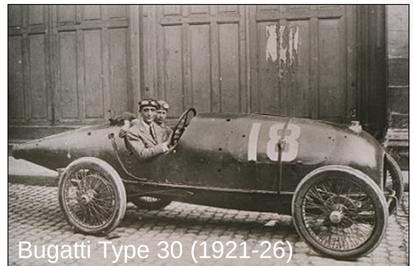
$$\begin{aligned} P_D &= F_D V \\ &= C_D A_f \frac{1}{2}\rho V^2 V \\ &= C_D A_f \frac{1}{2}\rho V^3 \end{aligned}$$

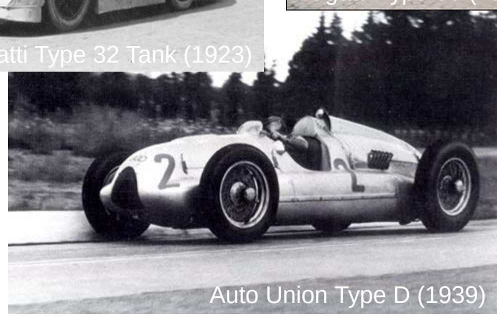
Fall 2015

2

Early Attempts


Bugatti Type 32 Tank (1923)

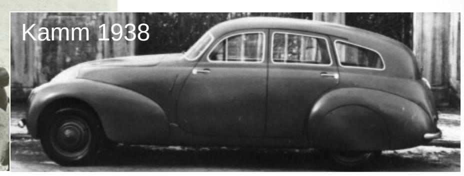

Bugatti Type 30 (1921-26)


Auto Union Type D (1939)

Fall 2015

Early Attempts (2)


Tatra T97 (1936)


Kamm 1938

Chrysler Airflow (1934)

Fall 2015

Early Attempts (3)

Current 'State of the Art'

Limitations on Aerodynamic Drag Reduction

- Styling

(chirp, chirp)

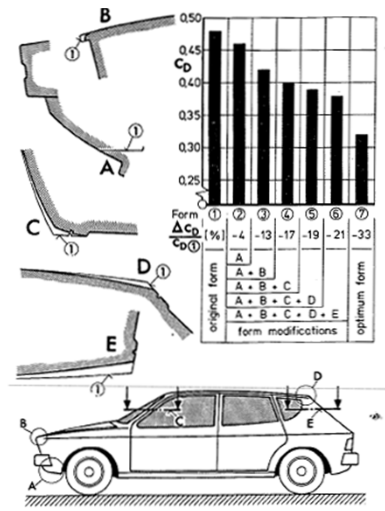
- Interior space
  - Total volume
  - Usefulness of available space
- Integration of automobile systems and components
  - Tires
  - Cooling
  - Ventilation
- Others?

The Optimization of Body Details – A Method for Reducing the Aerodynamic Drag of Road Vehicles

- Authored by W. H. Hucho, L. J. Janssen and H. J. Emmelmann of Volkswagen AG (Germany)
- SAE 760185

- Detailed an approach where the details of the aerodynamic configuration were adjusted to improve performance (drag reduction, lift reduction) without significantly changing the style or envelope of the vehicle

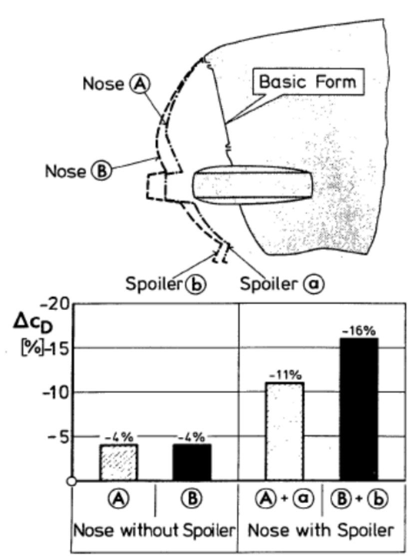
‘Step by Step’ Drag Optimization



Fall 2015

9

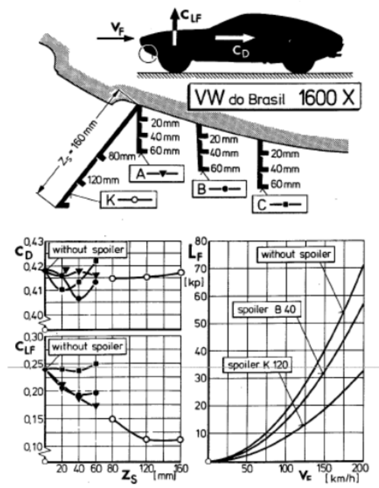
Most Aerodynamic Components Interact



Fall 2015

10

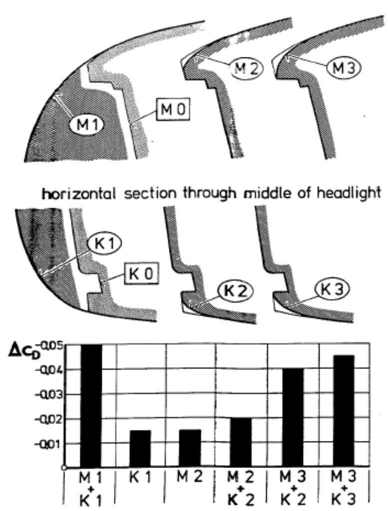
Effect of a Front Spoiler



Fall 2015

11

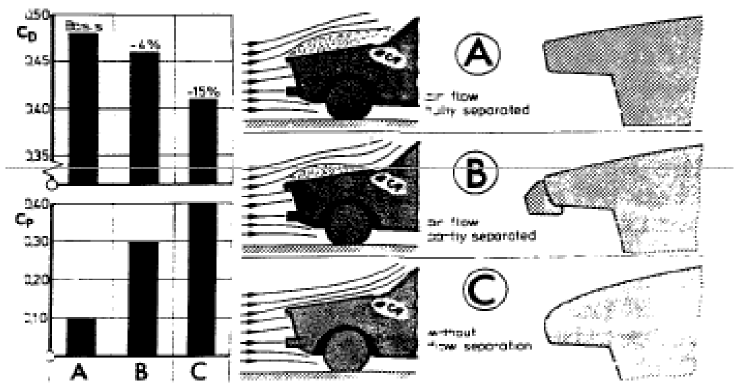
Leading Edge Details



Fall 2015

12

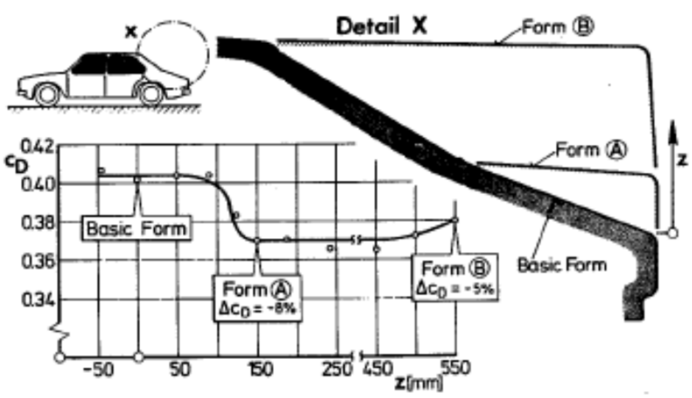
Ventilation Intake Pressure



Fall 2015

13

Notchback Elements



Fall 2015

14

Rear Sidepanel Taper

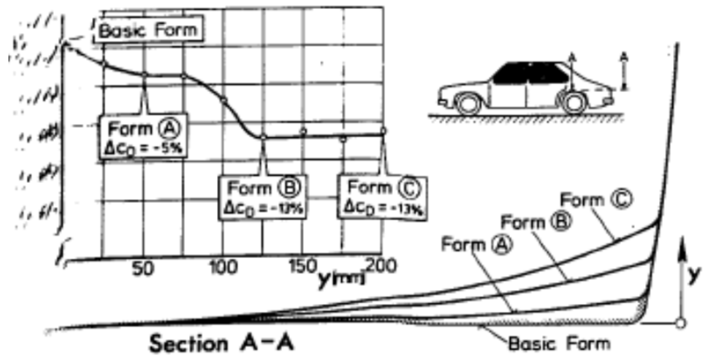
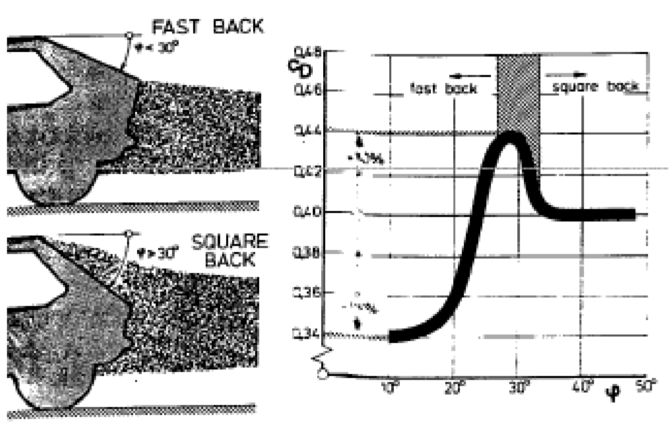


Fig.23 -Air drag versus side panel contraction y

Fall 2015

15

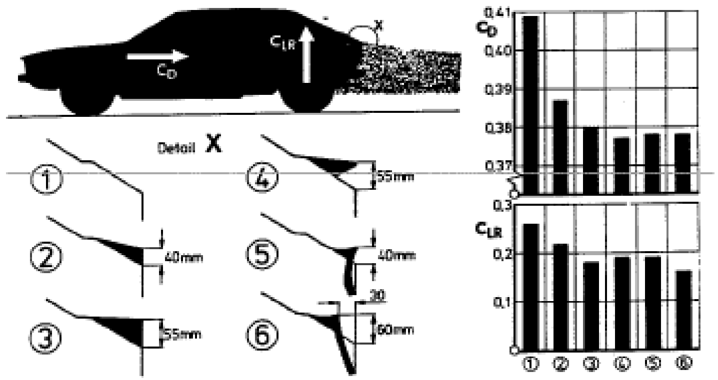
Hatchback Angle



Fall 2015

16

Rear Spoiler

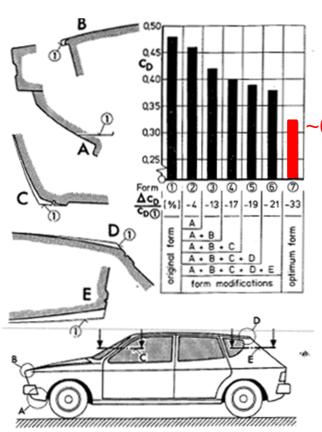


Fall 2015

17

Recall…

‘Step by Step’ Optimization



What about current modern cars?

- 0.29 – Porsche 996 GT3
- 0.35 – Ford GT
- 0.31 – Volkswagen Golf GTI 02
- 0.36 – Pagani Zonda F
- 0.35 – Koenigsegg CCR
- 0.30 – Audi S4 Avant 2003

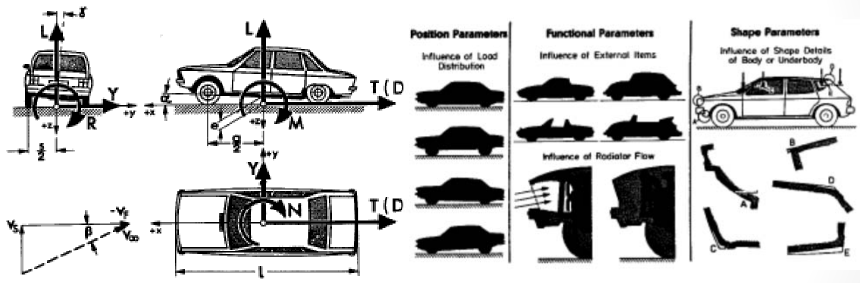
Fall 2015

18

C_D Values from Sport Auto Supertest

0.27 – Porsche 997 Carrera S
0.28 - BMW 335i
0.28 - Porsche Cayman S
0.29 - Porsche 987 Boxster S
0.29 – BMW E39 M5]
0.29 – Porsche 997 GT3
0.29 – Mercedes-Benz SL500 02
0.29 – Mercedes-Benz C32 AMG
0.29 – Porsche 996 GT3
0.29 – Chevrolet Corvette C6
0.29 – Porsche 996 Carrera 2
0.29 – Porsche Boxster 99
0.29 – Mercedes-Benz C55 AMG
0.30 – Porsche 996 GT3RS
0.30 – Audi S4 Avant 2003
0.30 – Porsche 996 GT3 MK2
0.30 – BMW E60 M5
0.30 - Porsche 997 GT3RS
0.30 – Chevrolet Corvette C5 CE
0.30 – Porsche 996 Turbo
0.31 – Audi S4 MK1
0.31 – Porsche 996 GT2 MK1
0.31 – Mercedes-Benz CLK55 00
0.31 – Mercedes-Benz SL55 AMG
0.31 – Callaway C12
0.31 – BMW M6
0.31- Volkswagen Golf GTI 02
0.31 – Ferrari 360 Challenge Stradale
0.31 – Mercedes-Benz SL65 AMG
0.31 – Audi RS6
0.31 – Audi B7 RS4
0.31 – Volkswagen R32 MK1
0.31- TechArt GT Street
0.32 - AC Schnitzer 6er Tension
0.32 – Honda Civic Type – R
0.32- Volkswagen Golf 4motion VR6 00
0.32 – Porsche 996 GT3 Cup
0.32 – Aston Martin DB7 GT
0.32 – Porsche Boxster S
0.32 –Volkswagen R32 MK2
0.32 – Ferrari 575M Maranello
0.32 – Nissan 350Z
0.32 – Mercedes-Benz C43 AMG
0.32 – Mercedes-Benz SLK55 AMG
0.32 – BMW E46 M3[/SIZE]
0.32 – Ferrari 575M Maranello
0.32 - Aston Martin DB9 Coupe
0.33 – Aston Martin V8 Vantage
0.33 – Aston Martin Vanquish
0.33 - Audi 2.0 TT TFSI
0.33 – Volkswagen Golf GTi DSG 05
0.33 – BMW E46 M3 CSL
0.33 – BMW M3 CSL
0.33 – Ferrari 550 Maranello
0.33 – Lamborghini Murcielago 6.2
0.33 – Aston Martin DB7 Vantage
0.33 – BMW Z4 3.0 CSi
0.33 – Audi RS4 MK1
0.33 – Alfa Romeo 156 GTA
0.33 – Aston Martin V8 Vantage
0.34 – Audi 1.8 TT MK1
0.34 - Lamborghini Gallardo Superleggera
0.34 – Ferrari F430 Berlinetta
0.34 – Maserati GranSport
0.34 – Honda NSX-R
0.34 – Mercedes-Benz CLK DTM
0.34 - MTM VW Golf V GTI vs
0.34 – Audi S3 MK1
0.34 – Jaguar XKR Coupe 01
0.35 – Lamborghini Gallardo
0.35 – Jaguar XKR, 2007
0.35 – Renault Megane Trophy Sport
0.35 – Mercedes-Benz SLK32 AMG
0.35 – Lamborghini Murcielago LP640
0.35 – Subaru Impreza WRX STi 04
0.35 – Subaru Impreza GT Turbo 00
0.35 - Mitsubishi EVO Carisma VI
0.35 – Lotus Esprit 350Z
0.35 – Koenigsegg CCR
0.35 – Ford GT
0.36 – Mini Cooper S Works
0.36 – BMW Z4 3.0 SMG
0.36 – Pagani Zonda F
0.36 – Alpina Roadster S
0.37 – BMW Z3 3.0 Coupe
0.37 – Lamborghini Diablo GT
0.37 - Steinmetz Opel Astra OPC
0.37 – Porsche Carrera GT
0.37 – Mitsubishi EVO Carisma VII
0.37 – BMW Z3M Coupe
0.38 – Opel Speedster Turbo
0.38 – Renault Clio V6 00
0.38 - Honda S2000 MK1
0.40 – AC Schnitzer V8 Topster
0.40 – Pagani Zonda C12S 7.3
0.42 – BMW Z8
0.42 – Lotus Exige S2
0.43 – BMW Z3M Roadster

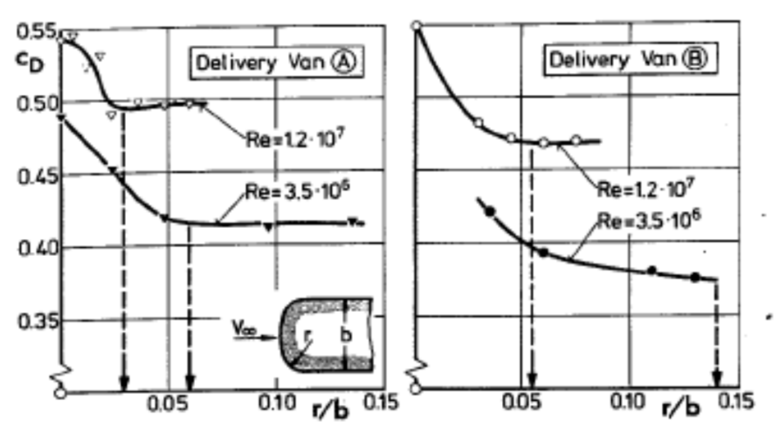
Backup Slides



Fall 2015

21

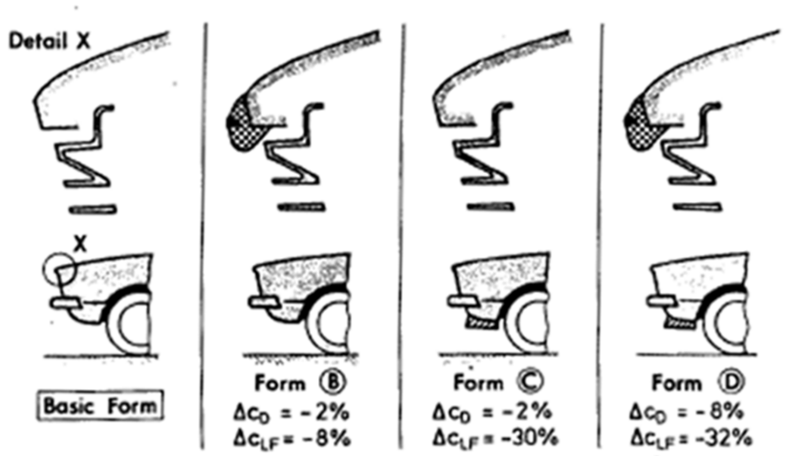
Reynolds Number Effects



Fall 2015

22

Interaction of Elements



Fall 2015

23